Analysis and Design of Algorithms

NUS National University of Singapore

Algorithms

CS3230

Tutorial

Week 13

Question 1

Given a simple undirected graph $G = (V, E)$, we would like to find the minimum vertex cover of the graph (i.e. find a subset of $V' \subseteq V$ such that for any $(u, v) \in E$, either $u \in V'$ or $v \in V'$).

Suppose we construct the vertex cover by greedily choosing the vertex that covers the most number of yet uncovered edges until all edges are covered. What is the best approximation ratio of the above greedy algorithm?

Hint: Consider the approximation algorithms covered in lectures.

- $O(\log^2 |V|)$
- $O(\log |V|)$
- It is an exact algorithm.
- Some constant α.

Question 1

Answer: B O(log |*V*|)

Reduce the problem to set cover.

- The set is the set of edges
- For each vertex, *v*, construct a subset corresponding to edges covered by *v*.

Set cover known to have approximation ratio of O(lg |*X*|), hence the algorithm has approximation ratio of O(lg |*E*|) = O(lg |V|).

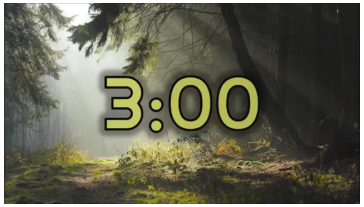
Question 2

Given a simple undirected graph $G = (V, E)$, we would like to find the minimum vertex cover of the graph (i.e. find a subset of $V' \subseteq V$ such that for any $(u, v) \in E$, either $u \in V'$ or $v \in V'$).

Instead of minimizing, it is often useful to ask the question of maximizing a related quantity. In this case, let's ask the question of *maximizing* the number of *edges* that we can find in E such that *no two edges share a vertex* in G. A set of edges where no two edges share a vertex in common is called a *matching* (*independent edge set*).

If we have a matching M of the graph G, what is the minimum number of vertices from V required to cover the edges in M?

- $|V|$
- $|E|$
- $2|E|$
- $|M|$
- $2|M|$

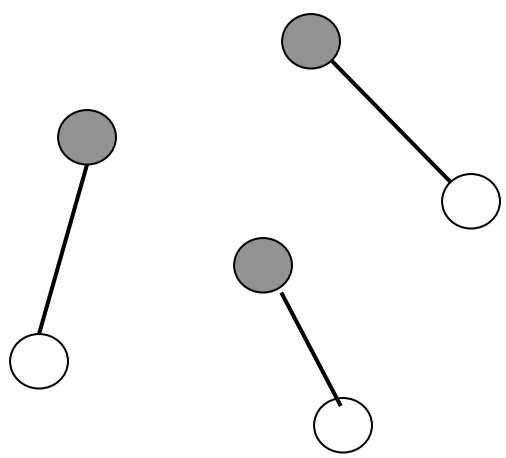


Question 2

Answer: D $|M|$

In a matching M, no two edges share the same vertex.

Hence, require exactly $|M|$ vertices to cover all the edges in M.



Matching of size 3

Cover of size 3

Question 3

Given a simple undirected graph $G = (V, E)$, we would like to find a *matching* in the graph, i.e. a set of edges M such that *no two edges share a vertex*.

Consider the following algorithm for finding M:

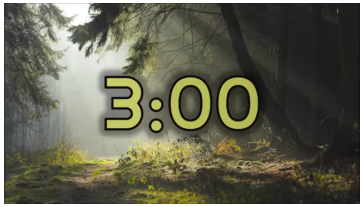
1. Select and arbitrary edge $(u, v) \in E$.
2. Add edge (u, v) to M and remove all edges covered by u and v from E.
3. If $E \neq \phi$, goto step 1.

- None of the statements.
- (I) only
- (II) only
- (I) and (II)

Let V_M be the set of vertices incident to the edges in M (all vertices connected to the edges) i.e. $V_M = \{v : \exists u \in V \text{ such that} \{u, v\} \in M\}$. Which of the following is true?

(I) The algorithm returns a **maximal** matching for the graph (it is not possible to add an edge to M and still have a matching).

(II) V_M is a vertex cover for G.



Question 3

Answer: D (I) and (II)
(I) is true.
Invariant: *M* contains a matching of removed edges and *E* does not share any vertex with *M*.
Initialization: True as start with empty *M* and no removed edges.

Question 3

Answer: D (I) and (II)

(I) is true.

Invariant: *M* contains a matching of removed edges and *E* does not share any vertex with *M*.

Maintainance: Assume true at start of iteration. Can add edge to *M* as *E* does not share vertex with *M*. Remove all edges in *E* that shares vertex with newly added edge, so *E* does not share vertex after iteration.

Question 3

Answer: D (I) and (II)

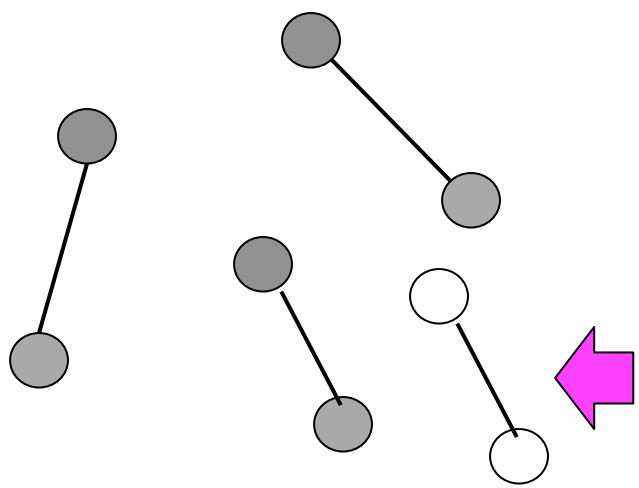
(I) is true.

We have argued that *M* is a matching when the algorithm terminates. To see that it is **maximal**, note that every edge removed during the iterations share a vertex with an edge in *M*.

Question 3

Answer: D (I) and (II)

(II) is true. V_M has all vertices connected to M and is **maximal**. Assume not a cover. Then can add edge (u,v) to M such that neither u or v is in V_M. Contradicts maximality of M.



New edge added, contradicting maximality.

Question 4

Given a simple undirected graph $G = (V, E)$, we would like to find a *matching* in the graph, i.e. a set of edges M such that *no two edges share a vertex*.

Consider the following algorithm for finding M:

1. Select and arbitrary edge $(u, v) \in E$.
2. Add edge (u, v) to M and remove all edges covered by u and v from E.
3. If $E \neq \phi$, goto step 1.

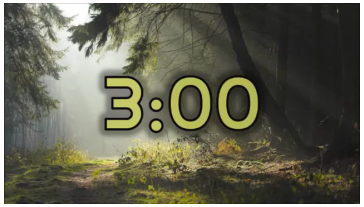
- None of the statements.
- (I) only
- (I) and (II) only
- (I) and (III) only

Let V_M be the set of vertices incident to the edges in M (all vertices connected to the edges) i.e. $V_M = \{v : \exists u \in V \text{ such that} \{u, v\} \in M\}$. Which of the following is true?

(I) The size of V_M is no more than twice the size of the minimum vertex cover.

(II) The algorithm for finding V_M is a 2-approximation algorithm for vertex cover.

(III) The algorithm cannot be a polynomial time algorithm as it solves the minimum vertex cover problem, and a polynomial time algorithm for solving the minimum vertex cover problem would imply P=NP.



Question 4

Answer: C (I) and (II)

(I) is true.

- Question 2 shows that size of optimal cover for M (hence for E, since M is subset of E) is at least $|M|$.
- Question 3 part (II) shows that the vertices incident to edges in M forms a vertex cover of size $2|M|$, giving approximation ratio of 2.

(II) restates (I) using definition of 2-approximation

Question 4

Answer: C (I) and (II)

(III) is false because the algorithm does not return the minimum vertex cover.

The algorithm runs in polynomial time but only gives a 2-approximation.